

Rahul Singha

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EDUCATION

Kalyani Government Engineering College <i>B.Tech in Computer Science and Engineering</i> • CGPA: 7.01/10 (through 4th semester) • Relevant Coursework: Data Structures & Algorithms · Design & Analysis of Algorithms Computer Organization & Architecture · Discrete Mathematics	Nadia, West Bengal 2024 – 2028
Amarkanandeshbandhu Vidyalaya <i>Higher Secondary (WBCHSE), Science – PCMB 80%</i>	Bankura, West Bengal 2021 – 2023
Amarkanandeshbandhu Vidyalaya <i>Secondary (WBBSE) 80%</i>	Bankura, West Bengal 2016 – 2021

TECHNICAL SKILLS

Languages: C, C++, Python, Dart, JavaScript, TypeScript
Frontend: HTML, CSS
Mobile: Flutter
Backend & Cloud: Firebase (Authentication, Cloud Firestore, Cloud Messaging, Realtime Database), Cloudflare Workers
Databases: SQLite
Tools: Git, GitHub, VS Code, Android Studio

PROJECTS

DakKhoj <i>Flutter, Dart, SQLite, Firebase Firestore, Firebase Auth, Workmanager</i> • Built and released an Android app for a postal office to track physical document records across pending, delivered, and archived states, with local SQLite as the primary source of truth • Designed an offline-first sync layer that queues additions, updates, and deletions offline and reconciles them with Cloud Firestore on reconnection • Scheduled background sync via Workmanager; integrated Google Sign-In to scope each user's cloud backup independently • Implemented local and cloud backup/restore to recover full document history after reinstallation GitHub	Aug 2026
Meal Poll App <i>Flutter, Dart, Firebase Auth, Firestore, FCM, Cloudflare Workers</i> • Designed a secure push notification workflow using a Cloudflare Worker as a server-side intermediary, keeping Firebase Admin credentials out of the distributed client • Enforced role-based access control across Admin, Manager, and Member roles through Firestore Security Rules as the primary enforcement layer • Built real-time meal response tracking with Google Sign-In and a member approval workflow; in use by 17–20 members GitHub	Jul 2026
WBJEE College Predictor <i>Next.js, TypeScript, Tailwind CSS, Firebase, Vercel</i> • Built a web app helping WBJEE candidates identify likely colleges and branches from previous-year cutoff trends, with rank-based prediction and Google Sign-In • Implemented filtering by category, quota, round, institute, branch, and district, with downloadable prediction reports • Deployed on Vercel and used by 200+ users GitHub	Jun 2026
Simple Weather Web <i>HTML, CSS, JavaScript, WeatherAPI</i> • Built a responsive web app displaying real-time weather by city, with error handling for invalid input GitHub	Jun 2025
GPT Clone <i>TypeScript, Gemini API, JavaScript, CSS</i> • Built a browser-based chat interface for interactive conversations with a generative AI model through the Gemini API GitHub	Apr 2025

HACKATHONS

ResQNet | *Kotlin, Jetpack Compose, JavaScript, Firebase Realtime Database, Leaflet.js, ESP32*

May 2026

- **ZYRO Hardware Hackathon**, Kalyani Government Engineering College
- Built the software layer of an accident-response prototype connecting an ESP32 sensor unit to hospital and police web portals through Firebase Realtime Database, with GPS-based incident location on Leaflet.js maps
- Developed separate dispatch workflows for hospital and police operators alongside a native Android companion app in Kotlin and Jetpack Compose | [GitHub](#)

PhaseSync | *Flutter, Dart, Provider, FL Chart*

Sep 2025

- **Smart India Hackathon 2025** (SIH25064, KSEBL) – College Internal Round
- Built a Flutter dashboard prototype for monitoring simulated solar, grid, and battery data with historical trend analytics, for a Kerala State Electricity Board problem statement | [GitHub](#)

CERTIFICATIONS

Internet of Things and AI Cloud Specialization – UC San Diego (Coursera), Mar 2024

Quantum Computing – C-DAC Hyderabad & IIT Roorkee, May 2025

Geospatial Analysis using Google Earth Engine – IIRS, ISRO, Feb 2024